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November 2, 2026

Quantitative Infrastructure Hiring Team
AI/ML Engineering Group
Tower Research Capital LLC
Gurugram, Haryana, India

Dear Quantitative Infrastructure Hiring Team,

Tower Research Capital's leadership in high-frequency algorithmic trading and low-latency infrastructure requires data pipelines that process massive event streams with predictable, microsecond-level performance. Engineering high-throughput feature retrieval under strict latency SLAs demands eliminating memory overhead and asynchronous queue contention rather than relying on brute-force hardware scaling. Having built and benchmarked local-first search and evaluation pipelines in **Python**, **C++**, and **Linux**, I have tackled the exact latency and concurrency bottlenecks that emerge when merging high-dimensional distributions under load.

In my open-source systems project **doubtless**, I developed an asynchronous hybrid retrieval engine fusing lexical BM25 algorithms with dense vector embeddings via Reciprocal Rank Fusion (RRF). By optimizing CPU instruction pipelines and binding custom **C++** routines to SQLite indexes, I achieved an empirical **86.9% Recall@5** at **8.9ms latency** on commodity CPU cores. Additionally, during my software development engineer internship at **LexoLegal**, I architected legal search APIs across **40,000 regulatory documents**, cutting query latency from **850ms to 110ms** through **Redis** distributed caching and indexed table partitioning.

My engineering philosophy prioritizes statistical rigor and empirical verification over speculative claims. In **ncert-rag-evals**, I built an automated evaluation framework encompassing **84 unit and integration test suites** using **pytest**, benchmarking context retrieval across **12,000 document chunks** to achieve **350 evaluations per second** with confirmed statistical significance (**$p < 0.01$**). I ensure every architectural tradeoff is verified against repeatable, deterministic test harnesses.

I am eager to apply this systems focus to the AI/ML Engineering Intern role at Tower Research Capital. I am available for immediate engagement, fully prepared to contribute in your Gurugram office, and welcome the opportunity to discuss how my systems and evaluation background can support your infrastructure initiatives.

Sincerely,

Aman Verma